

Lecture #19, Nov 18, 2025

Iterative method continue:

unique solution of $Au = f$, $A \in \mathbb{R}^{n \times n}$, $u \in \mathbb{R}^n$, $f \in \mathbb{R}^n$

If A is not singular, then $Au = f$ has a unique solⁿ

If A is symmetric, we look for $u \neq 0$ and $u^T A u = 0$

If this happens, A is singular.

A symmetric matrix A is not singular if $u^T A u \neq 0$ for any $u \neq 0$.

check: If A is symmetric, we have $A = \Phi^T \Lambda \Phi$

Then $Au = \Phi^T \Lambda \Phi u$

let $y = \Phi u$ then $y^T = (\Phi u)^T = u^T \Phi^T$

$$u^T A u = u^T \Phi^T \Lambda \Phi u = y^T \Lambda y = \sum_{i=1}^n$$

$$= \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$
$$= \begin{bmatrix} y_1 & y_2 & \cdots & y_n \end{bmatrix} \begin{bmatrix} \lambda_1 y_1 \\ \vdots \\ \lambda_n y_n \end{bmatrix}$$

$$= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2 = \sum_{i=1}^n \lambda_i y_i^2$$

So $u^T A u = 0$ is possible if all eigen values are zero.

Hence A is not singular if it is symmetric.

Remark: Aim to get symmetric A from F.E.M.

Eigen values by Fourier modes

let $A = [a_{ij}]$ with $a_{ii} = \alpha$, $a_{i, i+1} = \beta$, $a_{i-1, i} = \beta$
otherwise $a_{ij} = 0$

Eigen values of A are $\lambda_j = \alpha + 2\beta G \frac{\pi j}{n+1}$, $j=1, \dots, n$

Note that $Au = \sum_{j=1}^n a_{kj} u_j$

$\downarrow$ j -th col

j -th row

$\left[\begin{array}{cccc} \alpha & \beta & & \\ \beta & \alpha & \beta & \\ & & \ddots & \\ & & \beta & \alpha & \beta \\ & & & \ddots & \\ & & & & \ddots \end{array} \right]$

$\left[\begin{array}{c} u_1 \\ \vdots \\ u_{j-1} \\ u_j \\ u_{j+1} \\ \vdots \\ u_n \end{array} \right]$

j-th row $\Rightarrow \alpha u_{j-1} + \beta u_j + \gamma u_{j+1}$ (need to correct)
 switch α, β .

Take $Au = \lambda u$ if λ is an eigen value.

Hence $\alpha u_{j-1} + \beta u_j + \alpha u_{j+1} = \lambda_j u_j$

Take $u_j = e^{i\theta_j}$, $i = \sqrt{-1}$

$$\underline{\alpha} e^{i\theta(j-1)} + \underline{\beta} e^{i\theta j} + \underline{\alpha} e^{i\theta(j+1)} = \lambda_j e^{i\theta j}$$

$$\cancel{\beta} e^{-i\theta} + \cancel{\alpha} + \cancel{\alpha} e^{i\theta} = \lambda_j$$

$$\boxed{\lambda_j = \cancel{\alpha} + 2\cancel{\beta} \cos \theta}$$

From 1st row of A $\Rightarrow \alpha \beta \alpha \dots$ $\begin{bmatrix} u_0 \\ u_1 \\ u_2 \\ \vdots \end{bmatrix}$

$$u_0 \alpha = 0$$

Consider two conditions $u_0 = 0 \Rightarrow e^{i\theta \cdot 0} = 0$

$$u_{n+1} = 0 \Rightarrow e^{i\theta(n+1)} = 0$$

1st condition suggest $u_0 = \sin \theta j = 0$

So $u_{n+1} = \sin(n+1)\theta j = 0$

$$\Rightarrow (n+1)\theta = \pi j$$

$$\theta = \frac{\pi j}{n+1}$$

$$\boxed{\lambda_j = \alpha + 2\beta \cos \frac{\pi j}{n+1}}$$

Ex: $A = \begin{bmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{bmatrix}$

$$\boxed{\lambda_j = 4 + 2(-1) \cos \frac{\pi j}{3+1}}$$

Q.

$$u^{k+1} = G u^k + c$$

Is it convergent if I don't know $\rho(G)$

Lemma/Theorem

$$\text{let } A = M - N$$

$$u^{k+1} = M^{-1} N u^k + M^{-1} f$$

Show that if A and $M + M^T - A$ are symmetric and positive definite, the scheme converges.
(It is a sufficient condition)

Proof:

$$\text{We have } G = I - M^{-1} A$$

let λ be an eig. value of G
 w be eig. vector.

$$\text{want } \lambda < 1$$

$$(I - M^{-1} A) w = \lambda w$$

$$\Rightarrow M(I - M^{-1} A) w = \lambda M w$$

$$M w - A w = \lambda M w$$

$$M w - \lambda M w = A w$$

$$(1 - \lambda) M w = A w$$

$$w^T (1 - \lambda) M w = w^T A w$$

$$(1 - \lambda) w^T M w = w^T A w > 0, \quad A \text{ is SPD}$$

$$\Rightarrow \lambda \neq 1$$

$$\Rightarrow \frac{1}{1 - \lambda} w^T A w = w^T M w$$

Take transpose

$$\frac{1}{1 - \lambda} w^T A^T w = w^T M^T w$$

⊕

$$\frac{1}{1-\lambda} \omega^T A \omega = \omega^T M^T \omega, \quad A \text{ Symmetric} \Rightarrow A = A^T$$

$$(\dagger) \quad \frac{2}{1-\lambda} \omega^T A \omega = \omega^T M \omega + \omega^T M^T \omega$$

$$\begin{aligned} \frac{2}{1-\lambda} \omega^T A \omega - \omega^T A \omega &= \omega^T M \omega + \omega^T M^T \omega - \omega^T A \omega \\ &= \omega^T (M + M^T - A) \omega \end{aligned}$$

$$\frac{2}{1-\lambda} \omega^T A \omega = \omega^T (M + M^T - A) \omega$$

$\uparrow$ SPD given $\uparrow$ SPD require.

want $\frac{1+\lambda}{1-\lambda} > 0$

$$\Rightarrow \boxed{\lambda < 1}$$

Scheme Converges.

two basic scheme.

1) Jacobi method. $A = M - N$, $M = \text{diag}(A)$

Rewrite $A = D + L + U$

$\uparrow$ diag $\uparrow$ lower $\uparrow$ upper

or $M = D$, $N = -(L + U)$

$$\begin{aligned} G &= I - M^{-1}A = I - D^{-1}(D + L + U) = I - I - D^{-1}(L + U) \\ &= -D^{-1}(L + U) \end{aligned}$$

H.W. Show Jacobi method converges if A is SPD

H.W. $A = \begin{bmatrix} 4 & 2 & 3 \\ 3 & -5 & 2 \\ -2 & 3 & 8 \end{bmatrix}$ Find Jacobi iteration matrix and verify if it converges.

2) Gauss-Seidel

$$A = M - N = D + L + U$$

$$M = L + D$$

Iterative matrix $G = M^{-1}N = -(L+D)^{-1}U$

H.W.
$$A = \begin{pmatrix} 4 & -1 & 0 \\ -1 & 4 & -1 \\ 0 & -1 & 4 \end{pmatrix}$$

we know $\rho(G) < \|G\|_\infty$

using this, show Jacobi method converges

H.W. show Jacobi method converges if A is
diagonally dominant. (Leveque ch 4)

Lecture #20, Nov 20, 2025

Finite element method

Ref: A first course in the numerical analysis of differential equation by Arieh Iserles

ch 9 : finite element method.

libMesh is a C++ library to do FEM using parallel adaptive mesh.

First principle in FEM is to find solution from a finite dimensional subspace.

$$\text{let } u_h(x) = \sum_{j=0}^N c_j \phi_j(x)$$

$\{c_j\}_{j=0}^N$ is a set of coefficients

$\{\phi_j(x)\}_{j=0}^N$ is a set of basis functions

We look for "best approximation" of $u(x)$ by $u_h(x)$

least square minimization (Loss minimization)

Find u_h s.t. $\arg \min_{u_h \in V_h} \|u - u_h\|_2$

$$\text{let } u_h(x) = \sum_{j=0}^n c_j \phi_j(x), \quad x \in [0, 1]$$

Error (loss, defect)

$$\begin{aligned} F &= \int_0^1 (u(x) - u_h(x))(u(x) - u_h(x)) \, dx \\ &= \langle u - u_h, u - u_h \rangle \end{aligned}$$

$$= \int_0^1 (u - u_h)^2 dx$$

$$= \int_0^1 \left[u(x) - \sum_{j=0}^N c_j \phi_j(x) \right]^2 dx$$

Set $\frac{\partial F}{\partial c_i} = 0$

$$F = \int_0^1 \left([u(x)]^2 - 2u(x) \sum_{j=0}^N c_j \phi_j(x) + \sum_{i=0}^N \sum_{j=0}^N c_i \phi_i(x) c_j \phi_j(x) \right) dx$$

$$\left. \begin{aligned} (a+b)^2 &= aa+ab+ab+bb \\ (a_1+a_2)^2 &= a_1a_1+a_1a_2+a_2a_1+a_2a_2 \\ (\sum a_j)^2 &= a_1a_1+a_1a_2+a_2a_1+a_2a_2 \end{aligned} \right\}$$

$$\frac{\partial F}{\partial c_i} = \int_0^1 \left(0 - 2u(x) \phi_i(x) + \sum_{j=0}^N 2c_j \phi_i \phi_j(x) \right) dx = 0$$

$$\sum_{j=0}^N c_j \int_0^1 \phi_j \phi_i dx - \int_0^1 u(x) \phi_i(x) dx = 0, \quad i=0, 1, \dots, N$$

let $\langle \phi_i, \phi_j \rangle = \int_0^1 \phi_i \phi_j dx$

$$\begin{bmatrix} \langle \phi_0, \phi_0 \rangle & \langle \phi_0, \phi_1 \rangle & \dots & \langle \phi_0, \phi_N \rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle \phi_N, \phi_0 \rangle & \langle \phi_N, \phi_1 \rangle & \dots & \langle \phi_N, \phi_N \rangle \end{bmatrix} \begin{bmatrix} c_0 \\ \vdots \\ c_N \end{bmatrix} = \begin{bmatrix} \langle u, \phi_0 \rangle \\ \vdots \\ \langle u, \phi_N \rangle \end{bmatrix}$$

$$Ac = f \quad / \quad \sum_{j=0}^N c_j a(\phi_i, \phi_j) = \langle f, \phi_i \rangle$$

Here $a(u, v) = \int_0^1 uv dx$ is called bilinear form.

Remark: i) $a(\alpha_1 u_1 + \alpha_2 u_2, v) = \alpha_1 a(u_1, v) + \alpha_2 a(u_2, v)$

ii) matrix $A = [a(\phi_i, \phi_j)]$ is called mass matrix.

iii) By solving $Ac = f$ we can find $c = \{c_j\}_{j=0}^N$ and hence $u_h = \sum c_j \phi_j$ is found by known ϕ_j

Galerkin projection

$$\text{let } u_h(x) = \sum_{j=1}^n c_j \phi_j(x)$$

want to find u_h s.t. $\|u - u_h\|_2$ is minimized

$$\text{Take } F(u_h) = \int_0^1 (u - u_h)^2 dx$$

want variation of F to be zero.

Find $v_h(x)$ s.t.

$$\int_0^1 (u - u_h) v_h dx = 0, \quad \text{for every } v_h \in \text{span}\{\phi_j\}$$

$$\text{let } v_h = \sum_{i=1}^n \alpha_i \phi_i(x)$$

$$\int_0^1 (u - \sum_{j=1}^n c_j \phi_j) \sum_{i=1}^n \alpha_i \phi_i dx = 0$$

$$\sum_{i=1}^n \alpha_i \left[\int_0^1 u \phi_i dx - \sum_{j=1}^n c_j \int_0^1 \phi_j \phi_i dx \right] = 0$$

To construct v_h , we need $\alpha_i \neq 0$

$$\text{this is possible if } \int_0^1 u \phi_i dx - \sum_{j=1}^n c_j \int_0^1 \phi_j \phi_i dx = 0$$

$$\Rightarrow Ac = f \quad (\text{same as least square method})$$

Remark: least square and Galerkin methods lead to

$$\boxed{a(u_h, v) = \langle u, v \rangle}, \quad a(u, v) = \int_0^1 uv dx$$

$$\Rightarrow a\left(\sum_j c_j \phi_j, v\right) = \langle u, v \rangle \quad \text{for any } v$$

Ex: $u_h = \sum_{j=1}^h c_j \phi_j$

Show that $u_h^v = \sum_{j=1}^h c_j \sum_{i=1}^h c_i \phi_i \phi_j$

Ex: Take $u(x) = x^2$, $\{\phi_j\}_{j=0}^1 = \{1, x\}$
i.e. $\phi_0 = 1, \phi_1 = x$

Find $\{c_j\}$ and $u_h(x)$

Ans: $u_h(x) = x - \frac{1}{6} x^2$

Ex: $u(x) = \sin(\pi x)$, $x \in [0, 1]$

Let $u_h(x) = c_1 x + c_2 x^2 + c_3 x^3$, $\phi_1 = x, \phi_2 = x^2, \phi_3 = x^3$

i) Compute the mass matrix $a(\phi_i, \phi_j)$

ii) Solve $Ac = f = \begin{bmatrix} \langle u, \phi_1 \rangle \\ \langle u, \phi_2 \rangle \\ \langle u, \phi_3 \rangle \end{bmatrix}$

Hint $a(\phi_1, \phi_1) = \int_0^1 x x dx = \int_0^1 x^2 dx = \frac{1}{3} x^3 \Big|_0^1 = \frac{1}{3}$

Galerkin approximation of PDE

$-u''(x) = f(x)$, $x \in [0, 1]$

$u(0) = \alpha$

$u(1) = \beta$

Idea: $u_h(x) = \phi_0(x) + \sum_{j=1}^h c_j \phi_j(x)$

$\phi_0(0) = \alpha, \phi_0(1) = \beta$, $\phi_0(x)$ satisfies B.C.

$$\phi_j(0) = 0, \quad \phi_j(1) = 0, \quad \text{for } j = 1, 2, \dots, n$$

$\{\phi_j(x)\}_{j=1}^n$ satisfy zero boundary condition.

that means $u_n - \phi_0(x) \in \text{span} \{\phi_j\}_{j=1}^n$

